UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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PART A (80 MARKS)

QUESTION 1

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) State six facts about substitution ciphers as presented in the lecture, covering how plain-text is transformed, what serves as the key, the security principle involved, and how the Vigenere cipher differs from simple substitution.

(6 marks)

QUESTION 2

a) Explain how a rail fence transposition cipher encrypts and decrypts a message.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

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Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) [COMPUTED PLACEHOLDER - columnar_3x4, 2 marks]

(2 marks)

b) Compare the columnar transposition cipher with the double transposition variant, covering how each encrypts, what serves as the key, and how the two differ in output.

(6 marks)

QUESTION 4

a) State TWO (2) formal security models and the specific security concern each addresses.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

QUESTION 5

a) Describe two ways substitution is used in classical encryption.

(2 marks)

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b) Decrypt the following ciphertext using columnar transposition with a 4 x 6 matrix (4 rows, 6 columns). Put the ciphertext into the columns, read the plaintext out of the rows, and ignore the filler X at the end. The key is the keyword secret.
FOIENCXLRPYXOMTSEEEUCLRX

(6 marks)

QUESTION 6

a) State TWO (2) key sources of software insecurity.

(2 marks)

b) Describe six (6) aspects of Windows security mechanisms, covering their purpose, the entities they manage, and the risks arising from their complexity.

(6 marks)

QUESTION 7

a) Explain two characteristics of Trojan horse malware that distinguish it from other malicious programs.

(2 marks)

b) Describe six techniques by which malware infects systems or propagates itself, including how it hides its activity and how it spreads through user interaction.

(6 marks)

QUESTION 8

a) List TWO (2) stages involved in the biometric authentication process.

(2 marks)

b) Explain six (6) points about biometric mechanisms and technologies, covering how a biometric is defined, the categories of biometric applications, and the strengths and weaknesses of specific biometric technologies.

(6 marks)

QUESTION 9

a) Explain the access control list (ACL) and access control matrix (ACM) structures.

(2 marks)

b) Refer to your answer in (a). Explain the goals of access control and the role of the reference monitor, then describe two formal security models and the specific concern each addresses, and explain the ACL and ACM structures.

(6 marks)

QUESTION 10

a) Explain how a chain of trust is established in trusted computing.

(2 marks)

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b) Discuss the NGSCB architecture by describing the roles of its Nexus and Nexus Computing Agent components and how they relate to the TPM. Provide six distinct points.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A computing professional at a company handles personal information about customers. The professional must decide how to act, guided by morals, ethics and laws, and by the ACM code of ethics and professional conduct.

a) Discuss how morals, ethics and laws differ, and explain how the ACM code guides the professional's conduct in the scenario.

(5 marks)

b) Explain five ways the ACM code of ethics and professional conduct should shape the professional's behaviour when handling customer personal information in the scenario.

(5 marks)

QUESTION 2

A security analyst is reviewing an incident in which an attacker injected JavaScript through a user request into a web application, and separately a program spread itself across machines on the network. The analyst must classify each event by the class of threat it represents and by the security objective that is affected.

a) Explain how the classes of threats map to impacts on confidentiality, integrity and availability, giving one distinct impact for each security objective.

(5 marks)

b) Referring to the scenario, state five facts about the Samy worm that show how an XSS payload can propagate itself and what impact it had.

(5 marks)

END OF QUESTION PAPER

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